

Alex Davison

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Software developer with a strong background in mathematics.

EDUCATION

Stony Brook University

Stony Brook, New York

- **Bachelor of Science in Mathematics**, Minor in Computer Science Expected May 2027
- **Courses:** Formal Verification and Computer Proofs, Algorithms, Cryptography (Graduate), Number Theory, Programming Abstractions

SKILLS

- **Programming:** Python, TypeScript, C++
- **Tools:** NumPy, Matplotlib, Linux, Bash, Git, GitHub
- **Math:** Analysis, Formal Verification, Algorithm Design
- **Core Competencies:** Problem-solving, Collaboration, Project Management, Leadership
- **Languages:** English (Native), Mandarin (Fluent)

EXPERIENCE

Reprogramming

Software Developer (Contracted)

2024

- Designed and developed a JavaScript web application for Thales Academy (a network of 13 Pre-K–12 schools) to automate creation of math worksheets for elementary students.
- Ensured alignment with curricula by designing features that allow teachers to tailor worksheet creation parameters.

DataAnnotation

Freelance Software Developer

2023–25

- Worked on a variety of script and codebase revisions across multiple languages (Python, Java, JavaScript, C, C++).
- Provided thorough code reviews, troubleshooting approaches, and detailed explanations of revisions implemented.

Math Tutoring

Private Tutoring

2023–24

- Tutored students weekly on subjects ranging from elementary level math to AP calculus.
- Helped struggling students catch up with their peers, and motivated overwhelmed students.

Mathnasium

2025–now

- Collaborated with other tutors to teach and adapt personalized lesson plans for over 50 K–12 students.

PROJECTS

EOcross Solver

2022

- Invented an algorithm for Rubik's cube EOcross solving that balances computer optimization with human intuition.
- Algorithm was adopted by the CrystalCube project (crystalcuber.com), used for training by several top speedsolvers.

ChessDB Mobile

2023

- Developed a mobile-friendly JavaScript web application for exploring chess openings, integrated with real-time statistics and evaluation data using APIs from chessdb.cn and lichess.org.
- Project serves as the original codebase that the ChessPath project (chesspath.pro) was built on, with hundreds of users.

STUDENT ENGAGEMENT

HackNYU 2025

2025

- Put together and led a four-person team. Guided brainstorming and project planning sessions.
- Developed front-end user profile UI, and back-end database support in Python with SQLite.
- Coordinated the merging of all contributors' work in Git/GitHub and produced the finalized project.

United States of America Computing Olympiad – USACO Bronze

2017

Competitive Chess – 1st place US Junior Chess Congress 2020, Stony Brook vs. Yale Invitationals 2024

2016–24